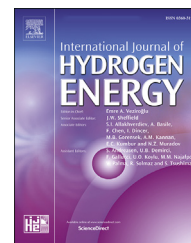


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Techno-enviro-economic analyses of hydrogen supply chains with an ASEAN case study

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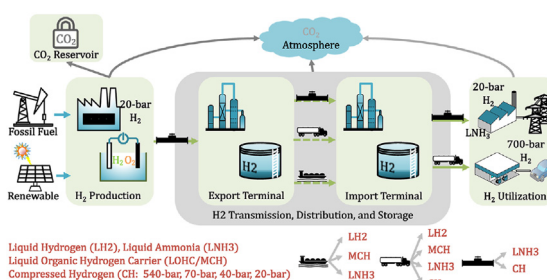
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HIGHLIGHTS

- Techno-enviro-economic analyses of H₂ transport via various carriers are presented.
- Sensitivity analysis of landed cost and carbon avoidance cost are discussed.
- Compressed H₂ via pipeline has the lowest energy penalty and cost for ≤2000 km.
- Liquid H₂ and methylcyclohexane as carriers have relatively high energy penalties.
- Progress on direct use of NH₃ can make it a preferred carrier for the power sector.

GRAPHICAL ABSTRACT



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ABSTRACT

Hydrogen is a promising low carbon fuel option with geographically distributed production and consumption. Hence, its regional and global hydrogen supply chains (HSCs) are vital for the potential future energy markets. We present a holistic study of various options for transporting (not producing) hydrogen from both techno-economic and environmental perspectives. The infrastructure and energy requirements of four options for transporting hydrogen between export and import terminals, namely methyl cyclohexane, liquid hydrogen, compressed hydrogen, and liquid ammonia, are analyzed in detail. These are compared for HSC energy penalty, carbon avoidance and landed cost of hydrogen under different scenarios. A case study is also presented to capture the perspectives of an importer. The preferred transport mode depends on export location and end use. For

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Transport
LOHC/MCH
Ammonia
Carbon avoidance

Singapore's power sector, compressed hydrogen from the neighbors via pipelines is most favorable with a carbon avoidance of 54–59% at 0.3 \$/kg CO₂ avoided.

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Nomenclature

CAPEX	capital cost, M\$
ACAPEX	annualized capital cost, M\$/y
FXOPEX	annual fixed operating cost, M\$/y
SFC	specific fixed cost, \$/kg
VC	variable operating cost, \$/kg
TPD	capacity, tonne/day
CH	compressed hydrogen
LH2	liquid hydrogen
MCH	methyl cyclohexane
TOL	toluene
LOHC	liquid organic hydrogen carrier
LNH3	liquid ammonia
HSC	hydrogen supply chain
SC	supply chain
EP	energy penalty
C	landed cost
A	carbon avoidance
AC	carbon avoidance cost

Introduction

The world is seeking scalable and affordable low carbon and/or renewable energy sources to reduce atmospheric CO₂ concentration while meeting the expected rising global energy demand. Notwithstanding significant advances in renewable technologies, today's energy scene is still heavily dependent on fossil fuels since most renewable energy sources are transient in nature and generally located far from the demand centers. Hydrogen is not only a potentially low-emission fuel that can be used across the transport, industry, and power sectors [1], but it can also bring renewable energy to the demand centers [2,3]. While its current large scale use is primarily limited to oil refineries [4,5] and ammonia production [2], hydrogen is receiving increasing attention worldwide as a versatile clean energy carrier for deep decarbonization. Low carbon hydrogen could be produced from a variety of technologies [6], such as green hydrogen from wind [7,8], hydro energy [9], and blue hydrogen from natural gas with CO₂ capture and sequestration [10]. There is a resurgence of interest in hydrogen economy around the world after the Paris Agreement [11]. Several countries (UK [12], USA [13], Australia [14], EU [15], China [16], etc.) and organizations (IEA or International Energy Agency [2], Hydrogen Council [17], etc.) have proposed region-specific hydrogen roadmaps to identify the

barriers for the transition to a sustainable hydrogen economy and strategies to overcome them.

While transport and storage are the key enablers for a flourishing hydrogen economy, the extremely low volumetric energy density of hydrogen and high energy penalties of compression and liquefaction of hydrogen are major barriers to its transport and storage enablers. Transport and storage are critical, because the abundant supply and major demand centers of renewable hydrogen are far apart and globally distributed. Comprehensive reviews of hydrogen transport and storage can be found in Wijayanta et al. [18] and Abe et al. [19]. Hydrogen can be stored as either a gas or a liquid. As shown in Table 1, its density at ambient conditions is extremely low: only 0.08 kg/m³ or 2.7 kWh/m³. Compressing hydrogen to 700 bar can increase this to 1332 kWh/m³, which is 1/10th of diesel. For a higher density, it must be liquefied at 20 K to store at atmospheric pressure, which is energy intensive. Even at these extreme physical conditions, its volumetric energy density is only 1/5th of diesel.

To overcome the above limitations of its direct physical storage, studying and assessing alternative storage options [20,21] are of utmost interest. These include storage in adsorbents (such as carbon-based materials and metal-organic frameworks/MOFs [22]), as single or complex metal hydrides [23] (such as LaNi₅H₆ and NaAlH₄) and in chemical compounds (such as ammonia, methanol and liquid organic hydrogen carriers or LOHCs). Among the emerging technologies for the large-scale, long-distance transport and storage of hydrogen, liquid hydrogen [24] (LH2), liquid organic hydrogen carriers [25] (LOHCs), and liquid ammonia (LNH3) are the expected forerunners with compressed H₂ (CH) as another option.

Extremely low temperatures, low energy efficiency, and high cost are the main barriers to the large scale implementation [26] of liquefaction. As with liquefied natural gas (LNG), management of the boil-off gas (BOG) becomes critical in controlling the storage pressure. BOG can be reduced via passive insulation [27] and recondensed by using a

Table 1 – Key properties of various hydrogen carriers.

Hydrogen Carrier	H ₂	CH	LH2	LOHC (MCH)	LNH3
T, K	298	298	20	298	240
P, bar	1	20–700	1	1	1
Density, kg/m ³	0.08	1.7–40	71	770	682
H ₂ , wt%	100	100	100	6.1	17.7
H ₂ Density, kg of H ₂ /m ³	0.08	1.7–40	71	47	121
Energy Density, kWh/kg	33.3	33.3	33.3	2.0	5.9
Energy Density, kWh/m ³	3	56.6–1332	2364	1565	4029
Energy Density Ratio vs H ₂	–	19–444	788	522	1343

liquefaction scheme. The state of the art in LH2 storage are the large tanks developed by The National Aeronautics and Space Administration (NASA) for the space program. These can keep the boil-off rate under 0.03% per day [28] in a one-million gallon spherical tank. In marine transport, BOG can also be used as a supplementary fuel [29] in the tanker engine. In contrast to LH2, LOHCs are an attractive option for hydrogen transport and storage at ambient conditions. The concept of storing hydrogen within LOHCs was proposed in the early 1980s. The term LOHC includes all hydrogen-rich and reversible forms of organic hydrocarbon liquids [30]. Some examples are Benzene – Cyclohexane [31], Toluene – Methyl Cyclohexane (TOL-MCH) [32,33], Dibenzyltoluene – Perhydrodibenzyltoluene [32], and N-ethylcarbazole – perhydro-N-ethylcarbazole [34]. In these pairs, the second compound is the hydrogenated form of the first. The TOL-MCH system has attracted much interest and has been demonstrated by a pilot-scale hydrogen supply chain (HSC) between Brunei and Japan [35]. Although its density per unit volume (see Table 1) is lower than LH2, MCH can be readily transported and stored in existing ships and storage facilities due to its physico-chemical similarities with petroleum fuels and petrochemicals. On the other hand, LNH3 is another promising hydrogen carrier [36], with a much higher hydrogen density than MCH. Although its transport requires 240 K and atmospheric pressure, the conditions are far milder than LH2. Furthermore, industrial ammonia production via the Haber-Bosch process is a mature technology in the fertilizer industry. LNH3 is not only a hydrogen carrier, but also a fuel substitute. It can be used directly for combustion [37,38] and also potentially as a marine bunker fuel.

LH2, LOHC, and LNH3 have been studied in the past decades. Kawasaki Heavy Industries Ltd. (KHI) Japan introduced the concept of a CO₂-free HSC in 2014 for blue hydrogen from Australia by coal gasification and carbon capture and storage (CCS) [29]. They estimated that liquefaction would account for 33% of the total cost. Cardella et al. [39] worked on technologies to reduce the liquefaction cost by 67%, while achieving a specific energy consumption of 5.9–6.6 kWh per kg LH2. Recently, Heuser et al. [40] also carried out a techno-economic analysis of a potential CO₂-free HSC via LH2 between Japan and Patagonia with green H₂ from wind-based water electrolysis. Teichmann et al. [33] compared the long-distance transport of energy via LOHC, LH2, and High Voltage Direct Current (HVDC). They demonstrated that importing green hydrogen as LOHCs from countries (e.g. Iceland, Spain, etc.) endowed with great renewable energy resources can be economically feasible compared to H₂ production via steam methane reforming (SMR). Niermann et al. [30] proposed and studied a storage and transport scheme based on various LOHCs, via a comprehensive process simulation. They used Aspen Plus to estimate the energy efficiencies of various options. They showed that the LOHC route can be an efficient and economically promising option for transporting and storing hydrogen. Brigljević et al. [41] studied four LOHC systems via detailed process simulation and technoeconomic analyses. They compared the capital investments and H₂ prices for the four options under market uncertainty. Chemical processing and shipping were the largest contributors to the total

installed and operating costs. Recently, Kojima et al. [42] investigated a simple and low-cost system for MCH production from toluene and hydrogen using fluctuating renewable electric power. High purity of MCH was successfully produced. Lee et al. [43] studied the cost and risk of a hydrogen delivery infrastructure using liquid hydrogen and gaseous hydrogen. The results indicated that the gaseous and liquid hydrogen systems was superior in terms of cost and safety respectively. Miao et al. [44] compared long distance renewable hydrogen transmission via cables and pipelines. Renewable electricity was transmitted via cables and then converted to hydrogen in the first case, while in the second case renewable electricity was used to produce hydrogen at source and then transported via gas pipelines. For a 4000 km of transmission distance, renewable hydrogen cost for the two cases are 7 \$/kg and 9\$/kg respectively. All in all, many techno-economic analyses of various H₂ carriers are presented. However, to the best of our knowledge, there is no techno-enviro-economic analysis of hydrogen supply chains via CH, LH, MCH, and LNH3. That is the goal of this study.

In this study, we present a comprehensive analysis and comparison of select promising options (CH, LH2, MCH, and LNH3) for HSCs. Considering the worldwide drive towards decarbonization, it is timely and important. Past studies have largely ignored key factors such as landed costs, carbon avoidances, carbon avoidance costs, and carbon taxes. We estimate HSC energy penalty, carbon avoidance, landed cost, and carbon avoidance cost of H₂ as functions of several factors such as hydrogen price, grid electricity price, grid CO₂ emissions factor, transport distance, geographical locations, and others. We do explore one case study from the perspectives of importers, who would like to develop and leverage supply chains that minimize the landed cost and maximize the carbon avoidance.

Hydrogen supply chains (HSCs)

We begin by introducing the nomenclature used to describe various HSCs. The general structure of the nomenclature is as follows:

<Form of hydrogen carrier> <Value for pressure in bar>
<Transmission mode>

The < Form of hydrogen carrier> = CH (compressed hydrogen), LH2 (liquid hydrogen at 1 bar), MCH (methyl cyclohexane at 1 bar), or LNH3 (liquid ammonia at 1 bar).

The < Value for pressure in bar> = 20, 40, 70, 540, or 700 for CH.

Similarly, the <Transmission mode> = P, S or T to represent transmission via pipeline, ship (marine transport) or truck respectively.

Next, consider the HSC in Fig. 1. Its two principal entities are one export and one import terminals. The former is in a hydrogen producing/exporting region or country; and the latter is in a consuming/importing region or country. The HSC can be within national boundaries or international. Hydrogen entering the HSC can be of five grades [2]: black, gray, blue, and green. Black, brown, and gray H₂ are produced from bituminous coal, lignite and natural gas (NG) respectively with no CCS, and hence significant carbon emissions. Blue H₂ is

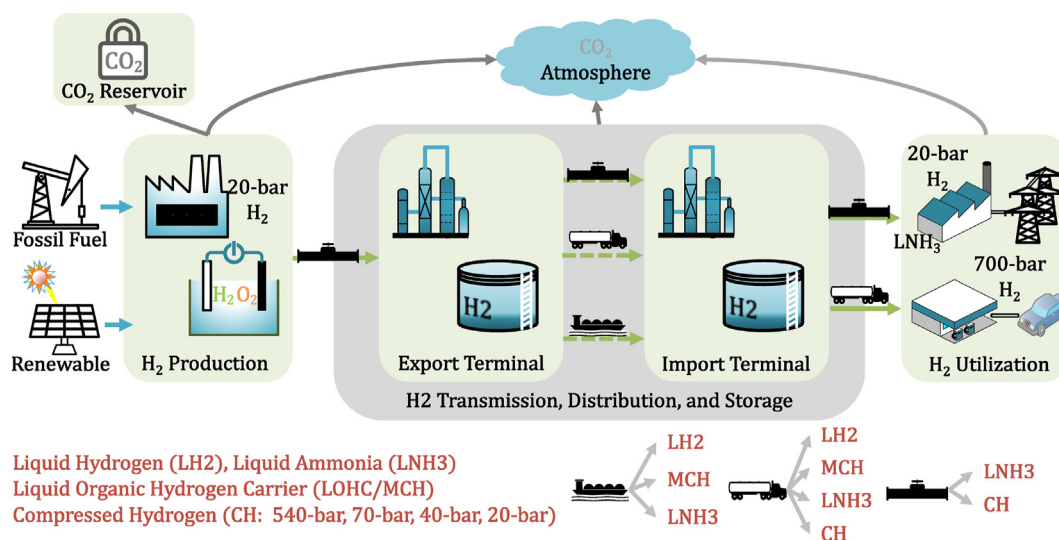


Fig. 1 – Structure of H₂ supply chain.

produced from a fossil fuel, but with CCS. Hence, its carbon emissions will depend on the CO₂ capture technologies. Finally, green H₂ is produced from fully renewable resources with zero carbon emissions.

Hydrogen received from the production plant is always assumed to come as compressed gas at 20 bar (CH20). For pipeline transmission between export-import terminals, CH20 is compressed to CH70 for delivering to the distribution network as CH40. Depending on the pipeline distance, repressurizing to CH70 may be required at intermediate booster stations. In each of the other HSCs, CH20 is first converted to LH2, MCH or LNH3 and stored at an export terminal. The carrier is then transmitted by ship to one or more import terminals (for example, LH2/MCH/LNH3 to LH2S/MCHS/LNH3S) where it is stored (LH2S/MCHS/LNH3S to LH2/MCH/LNH3). It is clear that storage tanks are necessary at export and import terminals for LH2, LNH3, and MCH. In addition, Toluene (TOL) storage tanks are also required at both terminals for an MCH supply chain. Either gaseous hydrogen is extracted from the stored carrier (for example, LH2/MCH/LNH3 to CH40) first for distribution or the stored carrier is distributed further before reverting to gaseous hydrogen (for example, LNH3 to LNH3P to LNH3). The carrier or gaseous hydrogen requires additional pressure conditioning to different levels depending on the end use in power/industry and transport sectors. Only LNH3 or gasified LNH3 can be directly used for combustion. Hydrogen fuel cell electric vehicles (FCEVs) represent the transport sector, CCGTs (Combined Cycle Gas Turbines) in the power plants represent the power sector, and stoves/burners represent the industry sector. The end users may receive hydrogen in four forms: CH20, LNH3, CH350, or CH700. The first two are for the power/industry sectors, and the last two are for FCEVs in the transport sector. Both pipeline and truck delivery options to the hydrogen refueling stations are considered for FCEVs (for example, CH40P to CH700/CH350 or LH2 to LH2T to CH700/350 or MCH to CH40P/CH540T to CH700/350, etc). The hydrogen

refueling stations, power stations, and industry/domestic users are the end nodes in the HSCs.

With four carriers (LH2, LNH3, MCH, and CH) and three delivery modes (pipeline, ship, and truck), there can be many HSC combinations. However, some are impractical or economically uncompetitive. For example, land transmission for LH2, MCH, and LNH3 may be uneconomical for long-distance inter-regional transmission between terminals. In this study, we focus on the nine HSCs listed in Table 2 where the forms of energy carriers at various points in the HSC, their modes of logistics, and the tasks that are performed in the HSC are clearly shown. For our discussion, comparisons, and analyses, we assign a unique HSC label to each such route. For example, we use SC700-LNH3S to signify that it delivers CH700 for end user consumption, while using LNH3S for transmission. Five HSCs (SC20-LH2S, SC20-MCHS, SC20-LNH3S, SC20-CH70P, and SCNH3-LNH3S) are suitable for the power/industry sector, and four HSCs (SC700-LH2S, SC700-MCHS, SC700-LNH3S, and SC700-CH70P) are suitable for the transport sector. We now define four criteria or metrics to assess and compare these HSCs.

Techno-enviro-economic metrics

Environmental impact, energetics, and economics are important attributes to assess the performance or attractiveness of an HSC. Hence, we define one assessment metric for each.

First, an HSC may need some energy at one or more steps. An HSC consuming excessive energy is undesirable, because a fossil fuel or some hydrogen will be required to generate that energy. The former will lead to carbon emissions, and the latter to hydrogen loss. We measure these two (fuel use and hydrogen loss) in terms of their LHVs and express this extra energy burden as the overall energy penalty *EP* (%) for the HSC as follows.

$$\text{Energy Penalty (EP, \%)} = 100 \times \frac{\text{LHV of feed CH}_2\text{OP (kWh)} - \text{LHV of H}_2 \text{ Delivered (kWh)} + \text{Energy (kWh) used by HSC}}{\text{LHV of H}_2 \text{ Delivered (kWh)}} \quad (1)$$

where, the LHV and HSC energy are for a unit amount of delivered H₂, and the numerator represents the net energy burden of the HSC for that amount.

Second, the commercial success of hydrogen demands favorable economics. Unless subsidized heavily by nations, a high-cost hydrogen cannot compete in a free market economy. Hence, we use the landed cost C (\$ per kg delivered H₂) as a key metric for an HSC.

Third, the ultimate goal is decarbonization (reducing CO₂ emissions). Using low emissions H₂ is meaningful, only if it reduces the total carbon emissions. To quantify or measure this reduction, also known as carbon avoidance, we must first select some bases or references that define the current practice. For the power sector, we take natural gas (NG) as the reference. For the transport sector, we take gasoline in the internal combustion engine of passenger vehicles as the reference. Then, we define carbon avoidance A (%) for an HSC as follows.

$$\text{Carbon Avoidance (A, \%)} = 100 \times \frac{\text{Emissions for NG or Gasoline} - \text{Emissions for H}_2}{\text{Emissions for NG or Gasoline}} \quad (2)$$

where, the emissions for H₂ and NG/Gasoline are for the same end goal, which can be either a unit energy or unit travel distance. The emissions for H₂ here include all emissions of its HSC from CH₂OP to end user consumption. The emissions for NG/Gasoline here include those from its supply chain and combustion. Since the use of pure hydrogen in CCGTs is not fully commercial yet, a reliable estimate of its power generation efficiency seems unavailable. Hence, we assume that H₂ and NH₃ have the same power generation efficiency as NG.

Finally, we combine the landed cost and carbon avoidance into a composite metric called carbon avoidance cost AC (\$/kg CO₂). This is nothing but the cost of avoidance via hydrogen as follows.

$$\text{Carbon Avoidance Cost (AC, \$/kg CO}_2\text{)} = \frac{\text{H}_2 \text{ Cost} - \text{NG or Gasoline Price}}{\text{Emissions for NG or Gasoline} - \text{Emissions for H}_2} \quad (3)$$

where, the costs and emissions for all four terms are for the same end goal as explained before.

Assessment metrics for HSCs

As the feed CH₂OP passes through an HSC, it undergoes conversion/storage at an import terminal; transmission, storage/reversion at an export terminal; distribution; and consumption at a power plant, industry site, or refueling station. Each of these steps consumes some energy, incurs some cost, and generates some emissions. By studying each step in detail, we estimate its (1) power use (kWh per kg of H₂), (2) hydrogen loss (kg per kg of H₂), (3) fuel use (kWh of MDO/diesel per kg of H₂), and (4) total cost (\$ per kg of H₂). Here, kg of H₂ refers to the hydrogen entering the step. Then, we use these to compute the contribution of each step to energy penalty (EP), carbon avoidance (A), and landed cost (C) for the entire HSC. Section C.1 of the Supporting Information has full details of our estimation procedures that make the following assumptions.

- 1) Pipelines transport CH₂OP from one or more production sites to the export terminal. The cost of transport is included in the price of CH₂OP.
- 2) Ships, barges, vessels, and tankers for distribution/transmission use a conventional fuel (e.g. marine diesel oil or MDO) with a known CO₂ emissions factor that consists of emissions from fuel combustion emissions and supply chain. Trucks use diesel with a known CO₂ emissions factor.
- 3) Process equipment (e.g. pumps and compressors) and operations (e.g. refrigeration and reliquefaction) use the electric power from a grid with a known CO₂ emissions factor.
- 4) Hydrogen, either from the feed or product, is used as a fuel for all heating needs (e.g. for dehydrogenating MCH, cracking NH₃, and separation) in each step, hence no additional carbon emissions are incurred.

- 5) Refrigeration cycles reliquefy any boil-off gases resulting from the storage and transfer of a cryogenic fluid (LH₂ or

Table 2 – Nine hydrogen supply chains via different carriers.

Source	Export Terminal		Transmit		Import Terminal		Distribute	End Node	Output	Use Sector	Label
	Convert	Carrier	Store ^a		Store ^a	Revert					
CH20P (Black, brown, gray, blue, or green)	Liquefy	LH2	LH2	Ship (LH2S)	LH2	–	Truck (LH2T)	Pressurize	CH700/CH350	Transport ^b	SC700-LH2S
						Gasify	Pipeline (CH40P)	–	CH20	Power/Industry	SC20-LH2S
	Convert and liquefy	LNH3	LNH3	Ship (LNH3S)	LNH3	Crack	Truck (CH540T)	Pressurize	CH700/CH350	Transport	SC700-LNH3S
						–	Pipeline (CH40P)	–	CH20	Power/Industry	SC20-LNH3S
	Hydrogenate	MCH	MCH & TOL	Ship (MCHS)	MCH & TOL	Dehydrogenate	Pipeline (LNH3P)	Gasify	LNH3	Transport	SCNH3-LNH3S
						–	Truck (CH540T)	Pressurize	CH700/CH350	Power/Industry	SC700-MCHS
	Pressurize	CH70	–	Pipeline (CH70P)	–	–	Pipeline (CH40P)	–	CH20	Transport	SC20-MCHS
							Pipeline (CH40P)	Pressurize	CH700/CH350	Power/Industry	SC700-CH70P
								–	CH20	Power/Industry	SC20-CH70P

^a Loading or unloading facilities are required for ship transmissions.^b Only CH700 is chosen for the analysis. Results for CH350 is not shown in this study.

LNH3), either onshore in a terminal or onboard a ship, hence no hydrogen losses from such operations. Boil-off gases from truck transport and delivery are nonzero and are counted as hydrogen losses.

- 6) Energy penalty in a step is given by the sum of the LHVs of all the primary energy fuels consumed by transport, and power generation and hydrogen losses. We use a power generation efficiency of 45% to convert power use into the primary LHV energy.
- 7) All costs are in 2019 USD. Capital cost of an entity (e.g. terminal) is first estimated for a USA installation. We then multiply this by a location factor [45] (e.g. USA = 1, Singapore = 1.2) to obtain the cost for another location.

We now compute the total cost contribution of each step.

Import/export terminals

In order to explain the costing methodology, let us consider a terminal receiving a feed at a given rate in TPD (tonne per day) of useable H₂. Table 2 lists all the steps taking place at each terminal. We compute the cost for each step as SFC (specific fixed cost) + VC (variable cost). SFC (\$ per kg H₂ in feed) is computed from the annual capital (ACAPEX) of the required equipment, their fixed operating costs (FXOPEX), and TPD. VC (\$ per kg H₂ in feed) is computed from the power consumption P (kWh per kg H₂ in feed) and hydrogen use/lost L (% of H₂ lost during operation/transfers or used as a fuel for heating needs. Section A of the Supporting Information details the procedures for the above steps using the parameters listed in Table 3.

Transmission and distribution

LH2S, MCHS, LNH3S, and CH70P are the potential carriers for transmission; and LH2T, LNH3P, CH40P, and CH540T are for distribution. Thus, we need to compute the cost (\$ per kg H₂) for ships, trucks, and pipelines.

For ships and trucks, we assume a specific dedicated physical asset (e.g. 100 K deadweight ton, dwt crude oil tanker for MCHS, 4-tonne truck for LH2) [2,46] for each carrier as listed in Sections B.1 and B.2 of the Supporting Information. Then, we compute two cost components, namely SFC and VC in \$ per kg H₂. SFC is simply the sum of annualized capital cost and annual fixed cost for the asset divided by the amount of carrier transported in one year. VC is simply the annual fuel cost (HDO or diesel) for the asset plus annual hydrogen loss (H₂ lost as a boil-off from the cargo tanks or used as a fuel in the ship) during transport. The details are given in Sections B.1 and B.2 of the Supporting Information. As expected, shipping cost is the highest for LH2S due to its specialized expensive ship, and the lowest for LNH3S due to its higher hydrogen carrying capacity per m³.

The piping cost for a carrier depends on the transport distance and flowrate. For CH40P and CH70P, we use the data and procedure from Hydrogen Delivery Scenario Analysis Model [47] by DOE, and use U.S. DOE's report [48] for LNH3P. Both of these take the delivery cost as SFC (capital cost and fixed operating cost for the pipeline) plus VC (the cost of power consumed by the compressor stations [49] and the cost of

Table 3 – Techno-economic data of processes involved in supply chains via LH2, MCH, NH3, and CH.

		P, kWh/kg	Loss, %	SFC, \$/kg
CH	CH20P – CH70P, compression of H ₂ from 20 bar to 70 bar	0.57	0.5%	0.025
	CH20P – CH40P, compression of H ₂ from 20 bar to 40 bar	0.32	0.5%	0.015
	CH20P – CH540T, compression of H ₂ from 20 bar to 540 bar	2.53	1%	0.96
LH2	CH20P – LH2, liquefaction of H ₂ received at 20 bar	10.0	0.5%	0.87
	LH2 – LH2S, loading liquid H ₂ to ships	0.008	–	0.07
	LH2S – LH2, unloading liquid H ₂ from ships to storage tanks (power supplied by ships)	–	–	0.07
	LH2 – LH2T, loading liquid H ₂ to trucks	0.004	–	0.03
	LH2 – CH40P, gasification of H ₂ to H ₂ at 40 bar	0.047	–	0.005
	LH2, 14-day, storage capacity as 14 times of the average daily terminal capacity	0.042	–	0.49
MCH	CH20P – MCH, hydrogenation of H ₂ received at 20 bar	0.017	–5.3%	0.09
	MCH – MCHS, unloading TOL from ships and loading MCH to ships	0.010	–	0.06
	MCHS – MCH, unloading MCH from ships to storage tanks and loading TOL to ships	0.009	–	0.06
	MCH – CH20P – CH40P, dehydrogenation of MCH to H ₂ at 20 bar	0.33	33.5%	0.21
	MCH – CH20P – CH540T, dehydrogenation of MCH to H ₂ at 20 bar	1.80	33.9	0.81
	MCH, 14-day, storage capacity as 14 times of the average daily terminal capacity	0	–	0.12
LNH3	CH20P – LNH3, liquid ammonia synthesis from H ₂ received at 20 bar	2.87	–	0.43
	LNH3 – LNH3S, loading liquid ammonia to ships	0.004	–	0.06
	LNH3S – LNH3, unloading liquid ammonia from ships to storage tanks	–	–	0.06
	LNH3 – CH20P – CH40P, cracking of liquid ammonia and compression of H ₂ to 40 bar	0.34	19.0%	0.19
	LNH3 – CH20P – CH540T, cracking of liquid ammonia and compression of H ₂ to 540 bar	2.19	19.4%	0.98
	LNH3 – LNH3P, pumping of liquid ammonia for pipeline distribution	0.007	–	0.002
	LNH3, 14-day, storage capacity as 14 times of the average daily terminal capacity	0.01	–	0.06
HRS	LH2T – CH700, refueling stations supplied by liquid H ₂ trucks	0.55	1.8%	2.11
	CH540T – CH700, refueling stations supplied by tube trucks with a storage pressure of 540 bar	1.76	0.5%	2.82
HRS	CH20P – CH700, refueling stations supplied by distribution pipelines	4.94	0.5%	3.32
	LH2T – CH350, refueling stations supplied by liquid H ₂ trucks	0.30	0.9%	0.88
	CH540T – CH350, refueling stations supplied by tube trucks with a storage pressure of 540 bar	0.65	0.5%	1.01
	CH20P – CH350, refueling stations supplied by distribution pipelines	3.09	0.5%	1.29

“–” means negligible.

hydrogen lost during the transport). Detailed data are in [Section B.4](#) of the Supporting Information.

End-user: power/industry sector

For this sector, hydrogen is used as received so no additional pre-treatment cost is involved. LNH3 delivered to end users is expressed in terms of equivalent kg H₂. Based on the LHV of NH₃ (5.25 kWh/kg) and H₂ (33.3 kWh/kg), 6.34 kg NH₃ is equivalent to 1 kg H₂.

End-user: transport sector

As stated earlier, hydrogen refueling stations³⁶ are the end nodes in the transport sector. We assume that a hydrogen refueling station can receive hydrogen in three forms (CH540T, LH2T, or CH20P) and can dispense it in two forms (CH700 for light-duty FCEVs such as passenger cars, and CH350 for medium to heavy-duty FCEVs). The refueling stations pump and gasify LH2T to CH350/CH700, and compress CH20P/CH540T to CH350/CH700 for delivery to the FCEVs. The

cost of a hydrogen refueling station entity is computed by adding the costs of all components, e.g. storage tanks, compressors, etc. In this study, Hydrogen Delivery Scenario Analysis Model and Heavy-Duty Refueling Station Analysis Model from the US Department of Energy are adopted for costing each step. In addition, we consider the loss of H₂ during transfers and storage. The relevant process/cost parameters are also given in [Table 3](#) for both CH700 and CH350.

Impact of key factors on supply chain metrics

Many technological, environmental, and economic factors and parameters impact the four metrics (EP, A, C, and AC) for an HSC. Some of these (e.g. carrier forms such as LH2S, MCHS, and NH3S; energy needs of various processing and transformation steps, etc.) are internal to or inherent in the definition of a supply chain and do not change. Others such as transmission distance, CO₂ emissions from grid power, etc. are external to the supply chains, and can vary with location and evolve over time. In this section, we study how some

major external factors impact the relative attractiveness or competitiveness of the nine HSCs listed in Table 2. Since it is not possible to study each external parameter, we select and vary only a few for each metric and fix others at some representative values. Except the parameter being varied, the nominal values of the other parameters held constant, unless stated otherwise, are as follows: grid CO₂ emissions factor = 0.5 kg CO₂/kWh, grid electricity price = 0.1 \$/kWh, CH₂OP price = \$2/kg, CH₂OP CO₂ emissions factor = 2.8 kg CO₂/kg H₂, transmission distance = 2000 km, location factor = 1, and carbon tax = 0 \$/t. More details are in Section C.1 of the Supporting Information.

HSC energy penalty (EP)

Fig. 2 shows the impact of transmission distance on EP for the nine HSCs listed in Table 2. As expected, EP increases with transmission distance for each HSC. The rate of increase is greater for the HSCs (SC20–CH70P and SC700–CH70P) with gas carriers, as compression is costlier than pumping. For CH70P, we assume one compressor station every 160 km [49]. SC20–CH70P has the smallest EP for transmission within 4500 km. For the same range, SC700–CH70P has a lower EP than other SC700s (HSCs delivering hydrogen gas at 700 bar). While SC20–CH70P and SC700–CH70P have the smallest EP for distances below 4500 km, their adoption for longer distances may be limited due to various economical, geographical, and sociopolitical factors.

For the seven HSCs with liquid carriers (LH₂S, MCHS, and LNH₃S), EP (Fig. 2) increases in the order SCNH₃–LNH₃S < SC20–MCHS < SC20–LNH₃S < SC20–LH₂S < SC700–MCHS < SC700–LNH₃S < SC700–LH₂S for distances below 5000 km. The MCHS-based HSCs have a greater slope for EP vs distance. This is because MCH has a relatively lower hydrogen

density (kg H₂ per m³ MCH) and needs more fuel for transport. SC20–LH₂S has a higher EP than SC20–MCHS and SC20–LNH₃S due to its energy-intensive liquefaction. SC20s and SCNH₃ always have lower EPs than SC700s because H₂ is not compressed to 700 bar. SC20–LNH₃S and SC700–LNH₃S have higher EPs than SC20–MCHS and SC700–MCHS respectively, because LNH₃ synthesis requires high power.

Landed cost of hydrogen (C)

The landed cost of hydrogen for an HSC depends on several factors such as CH₂OP price, transmission/distribution distances, grid (electricity) price, HSC entity locations, carbon tax, fuel prices, and discount rate. We observed that fuel prices (diesel and MDO) and distribution distance had little impact on the total cost, so we fix them at some nominal values. To keep the factors in our study manageable, we fix the demand location and its grid price as 0.10 \$/kWh. Then, we study the effects on the landed H₂ cost of five factors in this section: transmission distance, CH₂OP price, carbon tax, grid electricity price, and location factor.

Fig. 3(a) shows the landed H₂ costs for the nine HSCs as a function of transmission distance. The costs are based on a CH₂OP price of \$2 per kg H₂. In contrast to all others, SC20–CH70P is affected strongly by transmission distance. It is the cheapest option only below about 2500 km. This is very similar to the case for PNG, which is less expensive [50] than LNG only below about 3000 miles. Above 2000 km, SCNH₃–LNH₃S is the cheapest HSC. SC20–LNH₃S and SC20–MCHS are the next cheapest HSCs that are very close to each other. As seen before, SC700s are always more expensive than SC20s (HSCs delivering hydrogen gas at 20 bar), except SC20–CH70P for distances above 10,000 km. Among the four SC700s, SC700–LH₂S is the cheapest by \$0.2–0.7/kg H₂ despite its high EP. This is simply because of the savings in compression costs. While MCHS and LNH₃S do not seem the best option for delivering CH700 within the investigated distances, they will eventually become cheaper than SC700–LH₂S at longer transmission distances due to the higher shipping cost of LH₂.

Fig. 3(b) shows the effect of CH₂OP price on the landed cost of hydrogen for the nine HSCs. As expected, SC20–CH70P followed by SCNH₃–LNH₃S gives the cheapest hydrogen irrespective of CH₂OP price. When SC20–CH70P is not an option, SCNH₃–LNH₃S seems the best HSC followed by SC20–LNH₃S and SC20–MCHS that are very close. This holds for SC700s as well. As seen in Fig. 3(a), their landed costs increase only slightly with transmission distance, hence SCNH₃–LNH₃S seems to be the best option for the power/industry sector. Interestingly, although SC20–MCHS and SC20–LNH₃S seem cheaper than SC20–LH₂S, SC20–LH₂S becomes cheaper after CH₂OP prices of 4.5 \$/kg and 6.0 \$/kg respectively. This is because SC20–MCHS and SC20–LNH₃S must use hydrogen for heating.

The impacts of carbon tax, location cost factor at the export terminal, and grid electricity price on the landed costs are discussed in Section C.3 of the Supporting Information. Carbon tax will have a greater impact on an HSC with smaller carbon avoidance. Location factor at the export terminal mostly impacts LH₂S-based HSCs, because liquefaction is much more capital-intensive than ammonia synthesis or toluene hydrogenation. This is due to the expensive

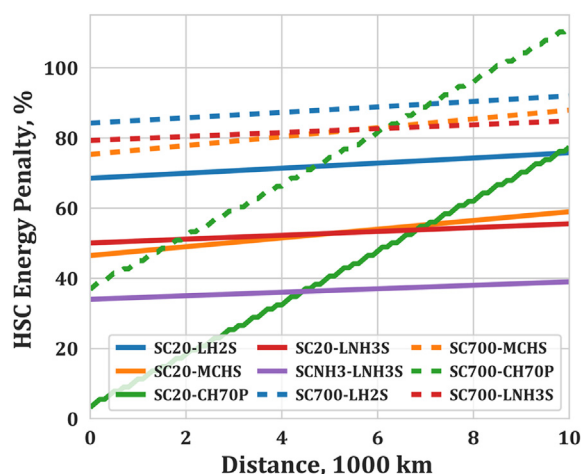


Fig. 2 – Hydrogen supply chain (HSC) energy penalty versus distance via different carriers for nine supply chains (Power/industry sector using CH₂O or NH₃: SC20–LH₂S, SC20–MCHS, SC20–CH70P, SC20–LNH₃S, and SCNH₃–LNH₃S; transport sector using CH700: SC700–LH₂S, MCHS, SC700–CH70P, and SC700–LNH₃S). Energy penalty increases as the distance increases, especially SC20–CH70P and SC700–CH70P.

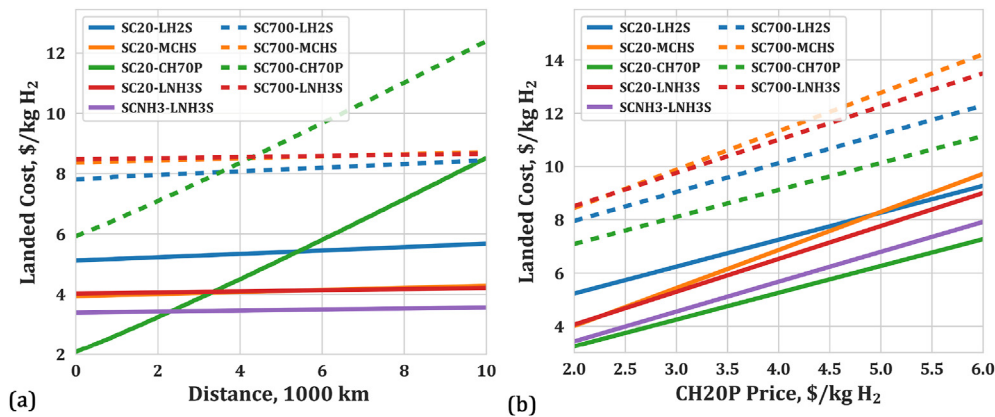


Fig. 3 – Landed cost versus transmission distance (a) and CH2OP price (b) via different carriers for nine supply chains (Power/industry sector using CH2O or NH3: SC20-LH2S, SC20-MCHS, SC20-CH70P, SC20-LNH3S, and SCNH3-LNH3S; transport sector using CH700: SC700-LH2S, MCHS, SC700-CH70P, and SC700-LNH3P). SC20/700-CH70P gives the lowest landed cost within 2000/3500 km for the end use in the power/transport sector, followed by SCNH3-LNH3S and SC700-LH2S who dominate long distance transmissions. SC20/700-MCHS is more sensitive to CH2OP price.

compressors and storage tanks required for LH2. Furthermore, the LH2S-based HSCs are also more sensitive to grid electricity price, as liquefaction needs power. SC20-CH70P and SC700-CH70P also need power, so are impacted strongly by the grid price and the effects become predominant at longer distances.

Finally, we examine the impact of SFC on the landed cost. Recall from Section [Assessment metrics for HSCs](#) that SFC has two terms: annualized capital cost and annual fixed operating cost. The latter is a fraction of the former, thus, SFC is proportional to the annual capital cost. Since capital cost estimation is generally approximate, we assume that its error is $N(0, 8\%)$, or normally distributed with zero mean and a standard deviation of 8%. Then, these accuracy estimates are used to establish 68% and 95% confidence levels on the landed costs for the nine HSCs as detailed in [Section C.4](#) of the Supporting Information. It can be seen that some confidence intervals overlap. For instance, SC20-MCHS (4.01 ± 0.08) and SC20-LNH3S (4.06 ± 0.10) may actually have the same landed cost. Similarly, SC700-MCHS (8.43 ± 0.50) and SC700-LNH3S (8.51 ± 0.50) may also have the same cost. Thus, LNH3-based and MCH-based HSCs are comparable in terms of the landed cost except SCNH3-LNH3S, where NH3 is used directly without cracking.

Carbon avoidance (A) and avoidance cost (AC)

We consider three factors in analyzing the nine HSCs in terms of their carbon avoidance and carbon avoidance cost. The first is the CO₂ emissions of the CH2OP source. This will reflect the technology used for producing hydrogen. The second is the grid CO₂ emissions factor. This will reflect the fuel mix used by the electricity. In future, most grids may use a mix of primary energy sources such as natural gas, biomass, solar, and wind. The third is the transmission distance. The nominal values of key system parameters and all the data used in computing avoidance and avoidance costs are listed in [Section C.2](#) of the Supporting Information.

[Fig. 4\(a\)](#) and [Fig. 4\(b\)](#) shows A and AC versus transmission distance for green CH2OP and a grid CO₂ emissions factor of 0.5 kg CO₂/kWh at the export and import terminals. While A shows a behavior very similar to EP in [Fig. 2](#), the type (heat vs power) of energy and its CO₂ emissions factor make a difference. For instance, MCHS-based and LNH3-based HSCs have greater heating than power needs. Due to the use of green CH2OP, they achieve higher A, but the price of green CH2OP affects their AC. In contrast, the HSCs (e.g. LH2S-based and LNH3-based) with greater power needs have lower A, as the grid power has nonzero emissions. Fuel for maritime transmission also impacts A, so A decreases with transmission distance for all ship-based HSCs. AC behaves similar to the landed cost ([Fig. 3\(a\)](#)) except that SC700s have lower AC than SC20s and SCNH3. SC700-CH70P has the lowest AC for transmission distances below 4000 km. For the power sector, SC20-CH70P seems the best option for distances within 3000 km, followed by SCNH3-LNH3S and SC20-MCHS. However, its AC surpasses SC20-LH2S for distances above 8000 km. Although SC20-MCHS has the highest A, its AC is higher than SCNH3-LNH3 due to its higher landed cost. Among the four HSCs with liquid carriers, SCNH3-LNH3S has the lowest AC, which still exceeds 1\$/kg CO₂. SC20-LH2S has the highest AC due of its high landed cost and carbon emissions. AC for SC700-LH2S, SC700-MCHS, and SC700-LNH3 are 0.3\$/kg CO₂ to 0.4\$/kg CO₂.

Similarly, the impact of transmission distance on A and AC for blue CH2OP is presented (see [Fig. C-7](#) of the Supporting Information). As expected, A is lower for all HSCs as compared to green CH2OP. This is markedly so for MCH-based and LNH3-based HSCs, because they must use hydrogen for their heating. The relative order of AC remains the same as for green CH2OP ([Fig. 4\(b\)](#)). Interestingly, AC is also lower for all SC700s and SC20-MCHS, SC20-LNH3, SCNH3-LNH3S, and SC20-CH70P for distances below 5000 km. This is because cheaper blue CH2OP results in lower landed costs. In fact, SC700-CH70P has a negative AC for distances within 2000 km

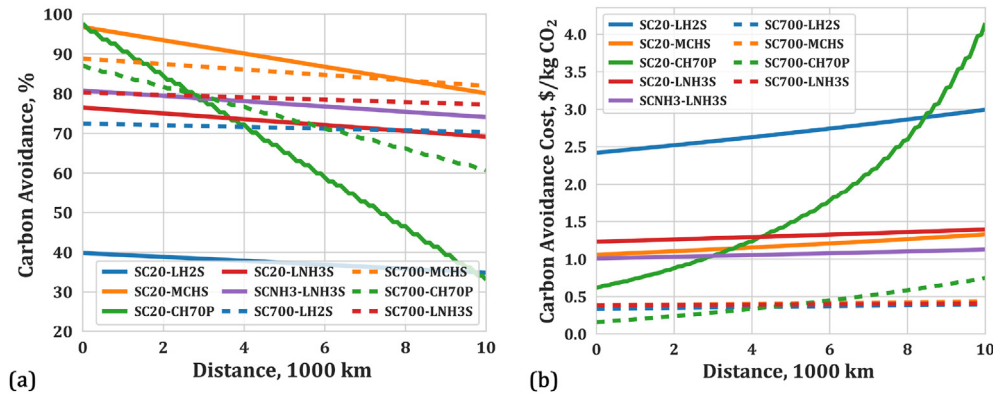


Fig. 4 – Carbon avoidance (a) and carbon avoidance cost (b) versus distance via different carriers for nine supply chains under green CH₂OP (Power/industry sector using CH₂O or NH₃: SC20-LH2S, SC20-MCHS, SC20-CH70P, SC20-LNH3S, and SCNH3-LNH3S; transport sector using CH₇OO: SC700-LH2S, MCHS, SC700-CH70P, and SC700-LNH3P). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

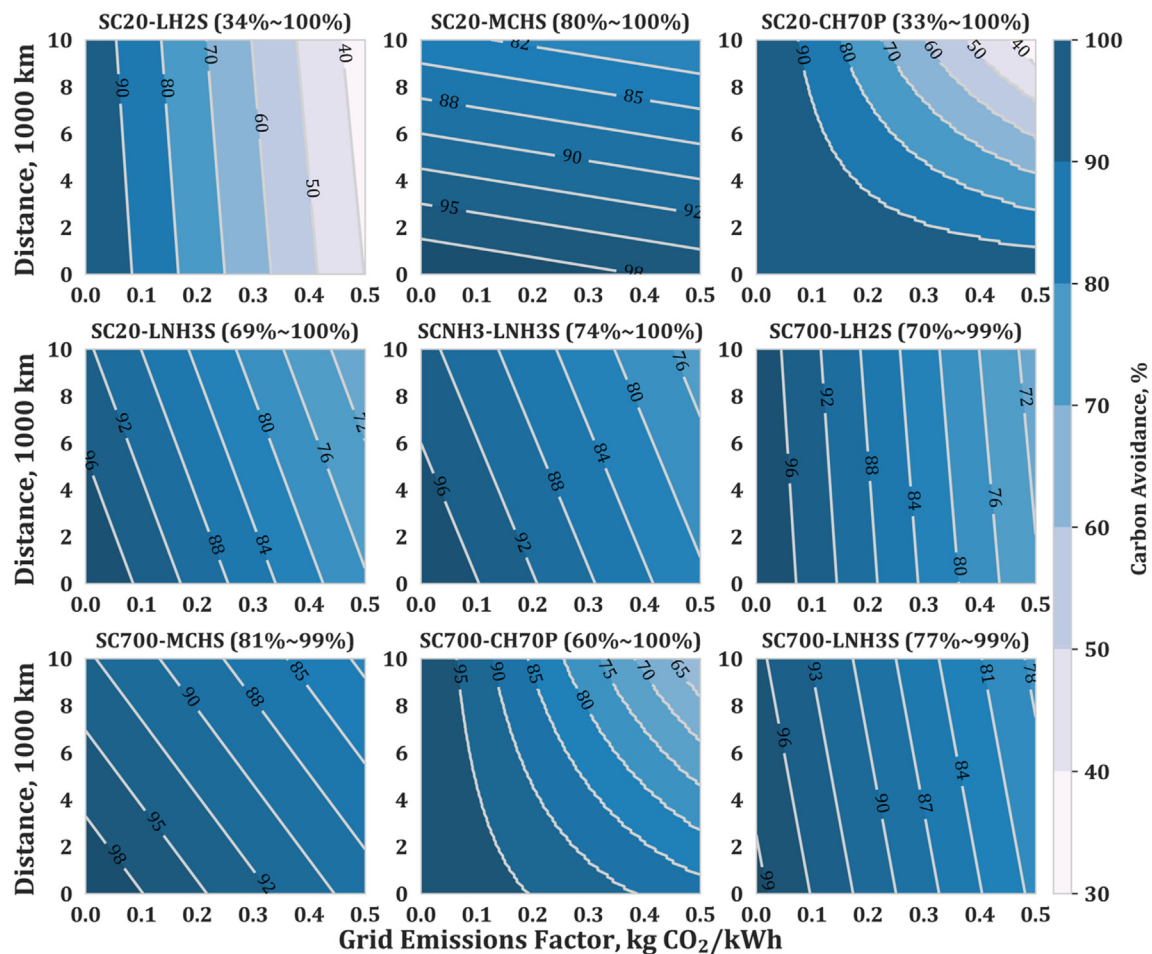


Fig. 5 – Carbon avoidance (%) heat map where color indicates carbon avoidance versus transmission distance (km) on y-axis and grid CO₂ emissions factor (kg CO₂/kWh) at both export and import sides on x-axis for nine HSCs with green CH₂OP as sources (Price decreases from 6 \$/kg H₂ to 2 \$/kg H₂ along with grid CO₂ emissions factor) with carbon avoidance cost range for each plot in parentheses. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

AC for SC700-LH2S, SC700-MCHS, and SC700-LNH3S is around 0.06–0.13 \$/kg CO₂. For SCNH3-LNH3, it reduces to 0.6 \$/kg CO₂ from 1 \$/kg CO₂ for green CH2OP. Since SC20-LH2S consumes high grid electricity with emissions, its A is below 10%, and AC is 7.5–66 \$/kg CO₂. AC for SC20-CH7OP reaches to 220 \$/kg CO₂ for distances above 10,000 km. The carbon avoidance even becomes negative.

Fig. 5 show the impacts of transmission distance, grid CO₂ emissions factor, and green CH2OP price on A as heat maps. For these figures, we have used the ranges of 2–6\$/kg for the green CH2OP price and 0–0.5 kg CO₂/kWh for the grid CO₂ emissions factor. The ranges of A are: 34–100% for SC20-LH2S, 80–100% for SC20-MCHS, 33–100% for SC20-CH7OP, 69–100% for SC20-LNH3S, 74–100% for SCNH3-LNH3S, 70–99% for SC700-LH2S, 81–99% for SC700-MCHS, 60–100% for SC700-CH7OP, and 77–99% for SC700-LNH3S. Unless the power grid has a zero CO₂ emissions factor, and the ships/trucks all run on renewable fuels, A is below 100% for any HSC.

A-range for the MCHS-based HSCs is smaller than the NH3-based HSCs. In other words, transmission distance and grid CO₂ emissions factor have relatively smaller impact on their A. However, the color gradient in the vertical direction shows that A decreases with the transmission distance. The ranges for LNH3S-based HSCs seem wider, because they need more power for ammonia synthesis and less energy for shipping and cracking. SCNH3-LNH3S has a higher A than SC20-LNH3S due to its lower heat and power demands.

The HSCs based on LH2S and CH7OP have much wider ranges than others. The sharper horizontal (vs vertical) gradient in color shade clearly shows the stronger impact of grid CO₂ emissions factor versus transmission distance on LH2S-based HSCs. This is because hydrogen liquefaction is energy-intensive and needs power from the grid. The results for SC20-CH7OP and SC700-CH7OP show that transmission distance and grid CO₂ emissions factor both have strong impacts, since power consumption increases with transmission distance. It can be expected that A will become negative for SC20-CH7OP and SC20-LH2S for higher grid CO₂ emissions factors. Hence, importing such a hydrogen makes no sense.

The impacts of the same three parameters on AC are shown in Fig. C-5 of the Supporting Information, whose patterns are similar to A in Fig. 5. The horizontal gradient

shows the strong impacts of CH2OP price and/or grid CO₂ emissions factor on all HSCs, while CH7OP-based HSCs show the impact of transmission distance. For the power/industry sector, AC for the four HSCs with liquid carriers drops to 0.3–0.5 \$/kg CO₂ for green CH2OP price below 3 \$/kg H₂ and grid CO₂ emissions factor below 0.1 kg CO₂/kWh. AC for SC20-CH7OP has a much wider variation. Its AC is minimal for distances below 2000 km, as shown in Fig. C-5 of the Supporting Information.

Fig. 6 shows the best (lowest AC) HSC option for various regions of grid CO₂ emissions factor or green CH2OP price and transmission distance, for both scenarios with and without CH7OP pipeline transmission as an option. The color intensity reflects the difference between the top two options. If pipeline transmission is allowed, then SCNH3-LNH3S seems the best option for longer distances. For shorter distances, SC20-CH7OP seems the best. However, if pipelines are not allowed, then SC20-CH7OP is fully replaced by SCNH3-LNH3S. Since shipping is a common mode of logistics in the ASEAN region, SCNH3-LNH3S seems a promising option.

For use in the transport sector, the range of AC for SC700-CH7OP seems the widest, as shown in Fig. C-5 of the Supporting Information. The three SC700s with liquid carriers have AC of 0.1–0.4 \$/kg CO₂. AC for SC700-CH7OP is negative for a small triangle below 2000 km and grid CO₂ emissions factor below 0.15 kg CO₂/kWh or CH2OP price below 3.2 \$/kg H₂. SC700-CH7OP is the preferred option for distances below 4000 km, as shown in Fig. 6. SC700-LH2S has the least AC, if SC700-CH7OP is excluded. But the light color shade indicates that the difference between the lowest AC and the second AC is mostly smaller than 0.05 \$/kg CO₂.

Case study

Considering the current lack of resources to produce hydrogen locally without importing natural gas or electricity, Singapore wishes to study options for importing hydrogen. Three potential hydrogen suppliers are considered: Malaysia (450 km, location factor = 0.96), Indonesia (950 km, location factor = 0.71), and Australia (5000 km, location factor = 0.97). Some supply chain alternatives are disallowed based on

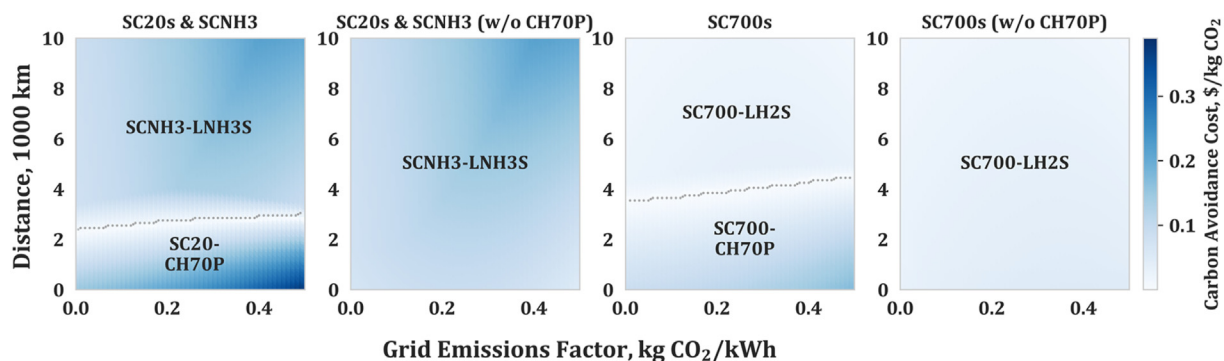


Fig. 6 – Preferred regions among SCNH3 and SC20s, and SC700s based on their carbon avoidance costs as functions of grid CO₂ emissions factor, green CH2OP price and transmission distance (The intensity of colors reflects the difference between the least and the second least carbon footprint options). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

exogenous geopolitical and techno-economic factors. For instance, an undersea pipeline from Australia to Singapore is deemed techno-economically nonviable. Considering its vast renewable resources, opportunities for carbon sequestration, and potential for producing and exporting green hydrogen, Australia can supply both green (6.0 \$/kg H₂) and blue (1.8 \$/kg H₂) hydrogen. Malaysia and Indonesia are endowed with natural gas resources, but they have limited renewable resources. Therefore, only blue H₂ from Malaysia (2.0 \$/kg H₂) and Indonesia (1.8 \$/kg H₂) is allowed. Then, we wish to answer the following questions:

From where and how much should Singapore import hydrogen?

What are the trade-offs among various import locations?

Which is the best HSC for the power/industry sector? Transport sector?

What are the landed costs of hydrogen for various HSCs?

What are the carbon avoidances and costs?

Landed cost (C)

Fig. 7 shows the landed costs for nine potential HSCs for Malaysia (M) and Indonesia (I), and fourteen potential HSCs for Australia (β for blue H₂ and α for green H₂). Solid bars represent the HSCs for the power/industry sector, and bars with dashed lines represent the HSCs for the transport sector. Blue bars are for LH2S, orange for MCHS, red for LNH3S, green for CH70P, and purple for direct use of LNH3. All assessment metrics are detailed in Section C.6 of the Supporting Information.

The landed costs for blue H₂ from Malaysia, Indonesia, and Australia vary between 2.3 \$/kg H₂ and 5.2 \$/kg H₂ for the power/industry sector, and 7.5 \$/kg and 9.2 \$/kg for the transport sector. SC20–CH70P (from Malaysia and Indonesia) has the lowest landed cost at 2.3–2.4 \$/kg H₂ followed by SCNH3–LNH3S from Indonesia at 2.9 \$/kg H₂. For the transport sector, SC700–LH2S has the second lowest landed cost after SC700–CH70P for any of the three sources. HSCs based on

MCHS and LNH3S are cheaper for Indonesia and Australia than Malaysia. This is because blue hydrogen is slightly more expensive in Malaysia due to higher NG price than Australia and higher capital cost than Indonesia. LH2S based blue hydrogen from Indonesia is also cheaper because of lower electricity cost. In case of Australia, despite a lower electricity cost, a long transmission distance outweighs the advantage leading to a higher landed cost for LH2S.

Since Green CH20P in Australia is three times more expensive than blue CH20P in any of the three countries, the landed costs are 1.5–3 times higher. For green hydrogen, SCNH3–LNH3 has the lowest landed cost (@8.0 \$/kg H₂) for the power/industry sector and SC700–LH2S has the lowest landed cost (@13.0 \$/kg H₂) for the transport sector. MCHS-based HSCs are more expensive than LNH3S-based HSCs, since they consume more hydrogen in the supply chain.

Carbon avoidance (A) and avoidance cost (AC)

Fig. 8 shows A and AC for all HSCs for the power/industry sector, represented by M for Malaysia, I for Indonesia, and β and α for blue and green H₂ from Australia. We observe that with green hydrogen from Australia, the lowest A (below 22%) is for SC20–LH2S (due to non-zero grid emissions) and the highest A (~90%) is for SC20–MCHS. While not shown in Fig. 8, the corresponding carbon avoidance via SC20–LH2S with blue hydrogen is negative from all three sources. This is because of their current high grid CO₂ emissions factors. A low A and a high landed cost result in a high AC (@4.7 \$/kg CO₂) for SC20–LH2S from Australian green H₂. Although the landed costs of SC20–LNH3S for blue H₂ (from all three countries) are less than half of those of SC20–MCHS, SC20–LNH3S, and SCNH3–LNH3S for green H₂ from Australia, these six HSCs have similar AC ~1.0–1.4\$/kg. Hence, SC20–MCHS seems the best option among them due to its highest A. If we consider HSCs with AC < 0.8 \$/kg CO₂, then the two SC20–CH70Ps from Malaysia and Indonesia seem the best. They both have a low AC ~0.3

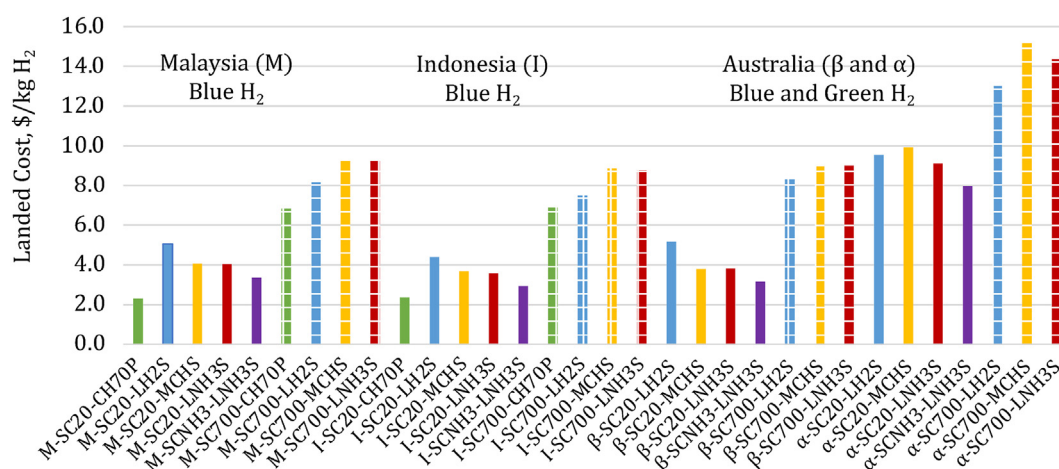


Fig. 7 – Landed cost for HSCs from Malaysia (M), Indonesia (I), and Australia (β for blue H₂ and α for green H₂) to Singapore (Blue, orange, red, green, and purple represent LH2S, MCHS, LNH3S, CH70P, and SCNH3–LNH3S). SC20/700–CH70P from Malaysia and Indonesia have the lowest landed cost, while the landed cost of green H₂ from Australia via all supply chains are higher than the one of blue H₂. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

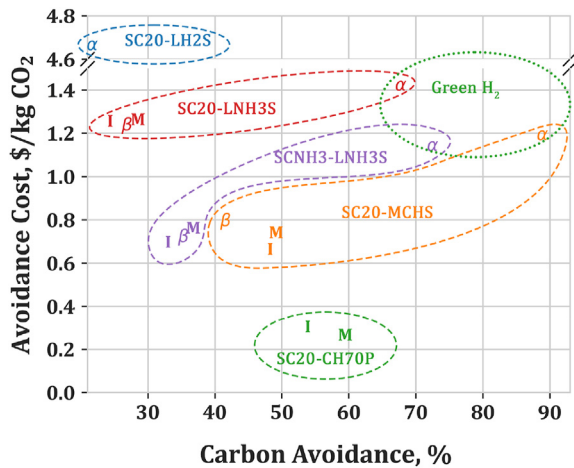


Fig. 8 – Carbon avoidance costs and carbon avoidances for SC20s and SCNH3 from Malaysia (M), Indonesia (I), and Australia (β for blue H₂ and α for green H₂) to Singapore (Blue, orange, red, green, and purple represent LH2S, MCHS, LNH3S, CH70P, and SCNH3-LNH3S). Blue H₂ via SC20-CH70P from Malaysia and Indonesia have the lowest carbon avoidance cost, while green H₂ via SC20-MCHS from Australia gives the highest carbon avoidance. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

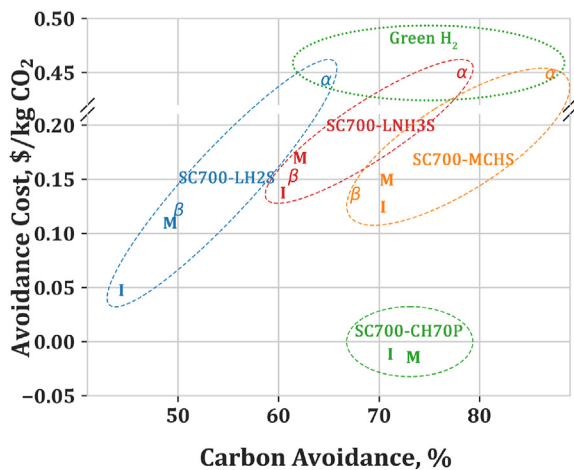


Fig. 9 – Carbon avoidance costs and carbon avoidances for SC700s from Malaysia (M), Indonesia (I), and Australia (β for blue H₂ and α for green H₂) to Singapore (Blue, orange, red, green, and purple represent LH2S, MCHS, LNH3S, CH70P, and SCNH3-LNH3S). Blue H₂ via SC700-CH70P from Malaysia and Indonesia have the lowest carbon avoidance cost, while green H₂ via SC700-MCHS from Australia gives the highest carbon avoidance. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

\$/kg CO₂, and A of 60% and 54% respectively. ACs for the six HSCs (SC20-MCHS and SCNH3-LNH3S) for blue H₂ from Malaysia, Indonesia and Australia are ~ 0.6–0.8 \$/kg CO₂ with AC < 50%.

For the transport sector (Fig. 9), all three HSCs from Australia for green H₂ have an AC ~0.4 \$/kg CO₂, while SC700-MCHS offers the highest A ~87%. In contrast, all the HSCs for blue H₂ are much cheaper with AC < 0.2 \$/kg CO₂. Among them, SC700-CH70P has the highest A and lowest AC, because it requires less power than SC700-LH2S and less hydrogen than SC700-MCHS and SC700-LNH3S. The HSCs via liquid carriers have comparable ACs, but A for SC700-MCHS is 10%–20% higher than SC700-LNH3S and SC700-LH2S.

In brief, SC20-CH70P (A: 54%–59% and AC: 0.3 \$/kg CO₂) and SC700-CH70P (A: 71%–73% and AC: 0.0 \$/kg CO₂) is the preferred supply chain to import from Indonesia and Malaysia. If a higher A > 80% is required, then SC20-MCHS and SC700-MCHS seem the best candidates for importing green hydrogen from Australia at the expense of a higher AC (1.2 \$/kg CO₂ for SC20-MCHS and 0.4 \$/kg CO₂ for SC700-MCHS). The higher A advantage of MCHS-based HSCs over NH3S-based HSCs is the higher electricity consumption by the latter.

Local carbon avoidance (A) and local carbon avoidance cost (AC)

The carbon avoidance and avoidance cost discussed did not account for the exact locations where emissions or costs occur. However, the accounting of emissions and costs is most often done for a specific location (e.g. country) or entity (e.g. enterprise). Hence, a country's or entity's policy makers may be interested only in the costs or emissions that matter to them. This motivates us to limit the focus of these two metrics to a local context. Thus, the A and AC (see Fig. C-11 of the Supporting Information) discussed in this section differ from their previous definitions as follows: (1) The CO₂ emissions factor for the reference NG/gasoline includes only the emissions from combustion (in other words, excludes the supply chain emissions) (2) Emissions occurring outside of Singapore (e.g. emissions from the production of hydrogen or any carrier outside Singapore), are not taken into account. In other words, the emissions from hydrogen sources (green or blue) and transmission are set to zero. Only power consumption at the import terminal and fuel consumption during truck distribution are included.

A exceeds 87% for all HSCs. HSCs with the same carrier and end user have same A. SC20-LH2S, SCNH3-LNH3S, and SC20-CH70P have ~100% A, as CO₂ from their negligible electricity use can be neglected. SC20-MCHS and SC20-LNH3S need some power in their compression stations for local distribution. Hence, they have 2–3% lower A than SC20-LH2S, SCNH3-LNH3S, and SC20-CH70P.

For the transport sector, compressing hydrogen to >700 bar requires power, so SC700-CH70P, SC700-MCHS, and SC700-LNH3S have A ~88%. Only SC20-LH2S has A ~98%, because it pumps LH2 to a high pressure before gasification, which consumes less energy than that for compression after gasification.

As H₂ sources do not impact A, an HSC with a lower landed cost will yield a lower local AC for the same carrier. As expected, α (Green H₂ from Australia) are above M (Malaysia), I (Indonesia), and β (blue H₂ from Australia). The differences between the HSCs for blue H₂ are very small, as the landed costs are comparable.

Based on local A and AC, CH70P-based HSCs from Malaysia and Indonesia are the best options, followed by SCNH3-LNH3S for the power/industry sector. SC700-LH2S is the best for the transport sector.

Remarks

The underlying assumptions made in this study contribute some degree of unavoidable uncertainty in the obtained results. For example, grid electricity emission factor has an impact on carbon avoidance, especially on supply chains with high electricity consumption. Power generation efficiency via H₂ or NH₃ are continuously improving as technologies are advancing. The impact of these uncertainties is addressed in [Section Landed cost of hydrogen \(C\)](#), where confidence intervals of the landed cost via various hydrogen carriers are shown. The observed overlaps indicate that there is no clear winner at this time and the pace of progress in the mentioned areas via further R&D will be critical determinants for the future.

Conclusions and future work

Nine hydrogen supply chains based on four carriers (LH2, MCH, LNH3, CH) for two end-use sectors (power/industry and fuel-cell based transport) were developed and compared in terms of landed cost, energy penalty, carbon avoidance, and carbon avoidance cost. Initial insights into hydrogen supply chains have resulted from this research. Pipeline transmission of 70 bar H₂ has the lowest HSC energy penalty for distances below 4500 km. LH2 and MCH as carriers have relatively high HSC energy penalties. However, the former can be economical for the transport sector. While MCH is the easiest among the three liquid carriers, further research on reducing its HSC energy penalty may enhance its competitiveness. LNH3 as a carrier, whether used as such or after being cracked back to H₂, is attractive in terms of the landed cost. Progress on utilizing it directly as NH3 may strengthen its case as the best carrier.

Based on a case study, we explored different options for Singapore and found importing blue H₂ via pipeline from neighboring countries gives the lowest landed costs (~2.4 \$/kg H₂ for the power/industry sector and ~6.9 \$/kg H₂ for transport sector) and carbon avoidance costs (~0.3 \$/kg CO₂ for the power/industry sector and ~0.0 \$/kg CO₂ for transport sector). For a higher carbon avoidance (>80%), Singapore should consider importing green H₂ via MCHS, but at the expense of a higher carbon avoidance cost. Further work to understand how robust of this solution is to variations in some of the assumptions will be forthcoming.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2021.07.138>.

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